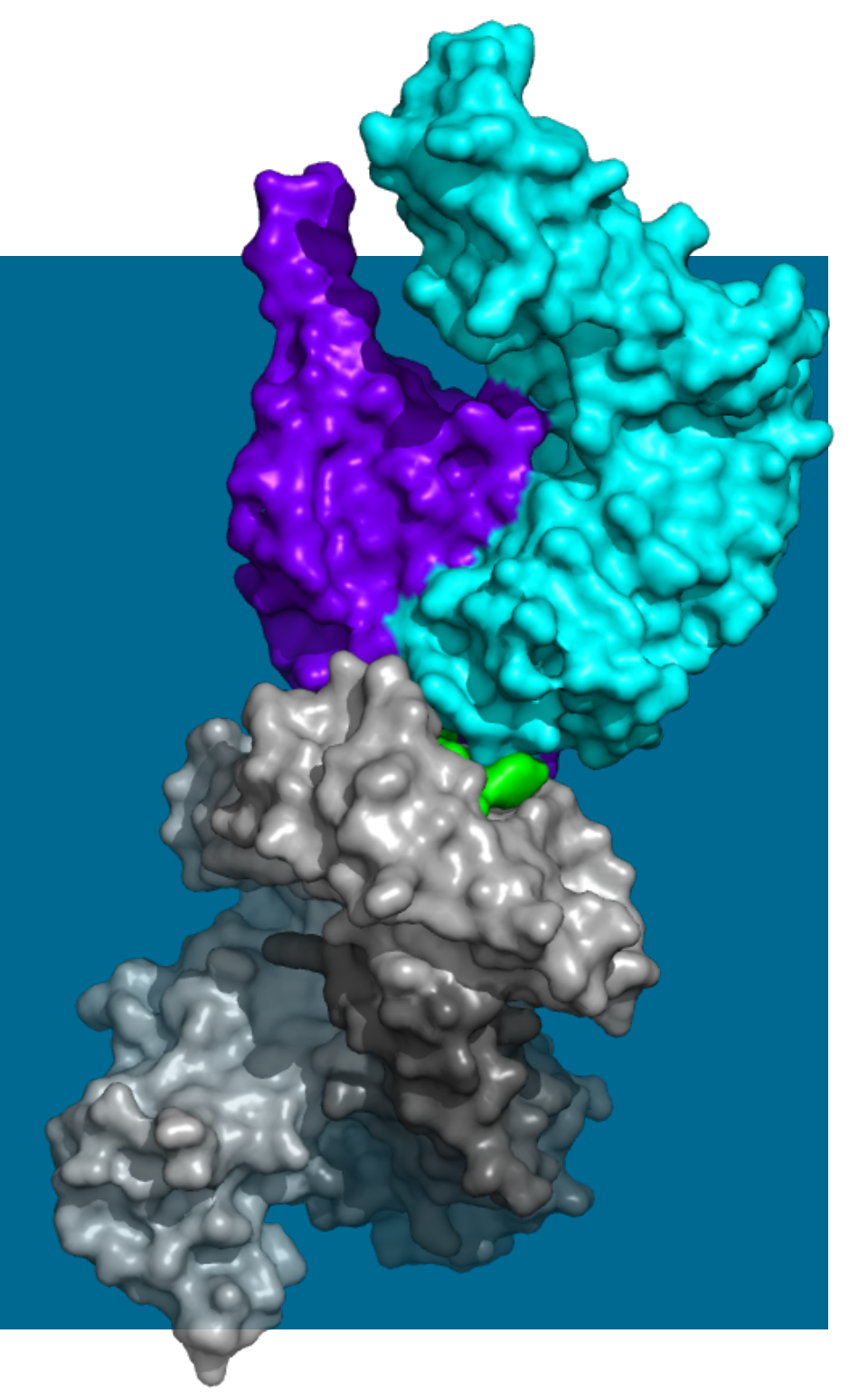


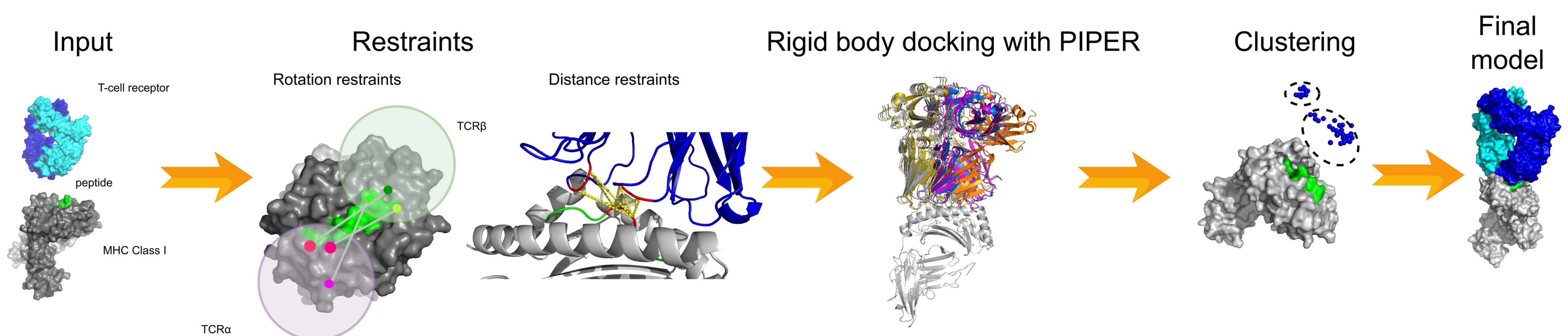
Efficient Computational Docking of TCR:pMHC Complexes Using Restricted Rotation Matrices and Distance Restraints



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Introduction

Understanding T-Cell Receptor (TCR) binding specificity to the peptide:MHC (pMHC) complex is crucial in immunotherapy and vaccine development. Computational docking is a valuable tool for modelling the 3D structures of complexes however the process can be computationally intensive. Here we present a computationally efficient rigid body docking algorithm designed to dock TCR and pMHC class I complexes. In this work we reduced the search space of PIPER[1] docking software with domain-knowledge-derived restraints and achieved state-of-the-art performances in very small computational time (< 5 mins per case on 30 CPU cores).

Objectives

- Increase docking speed for TCR:pMHC Class I complexes.
- Improve rigid body docking model quality

Methods

- Benchmark with experimental structures from docking benchmark dataset [2].
- Apply distance restraints to the CDR3 loops and the peptide.
- Generate rotation restraints based on known binding angles uniformly distributed on the globe (Figure 1) [3].
- Apply rotation restraints (Figure 1) [4].
- We select the top 1000 candidate complexes from billions of conformations sampled by PIPER.
- Cluster candidates based on interface-RMSD to select top models.

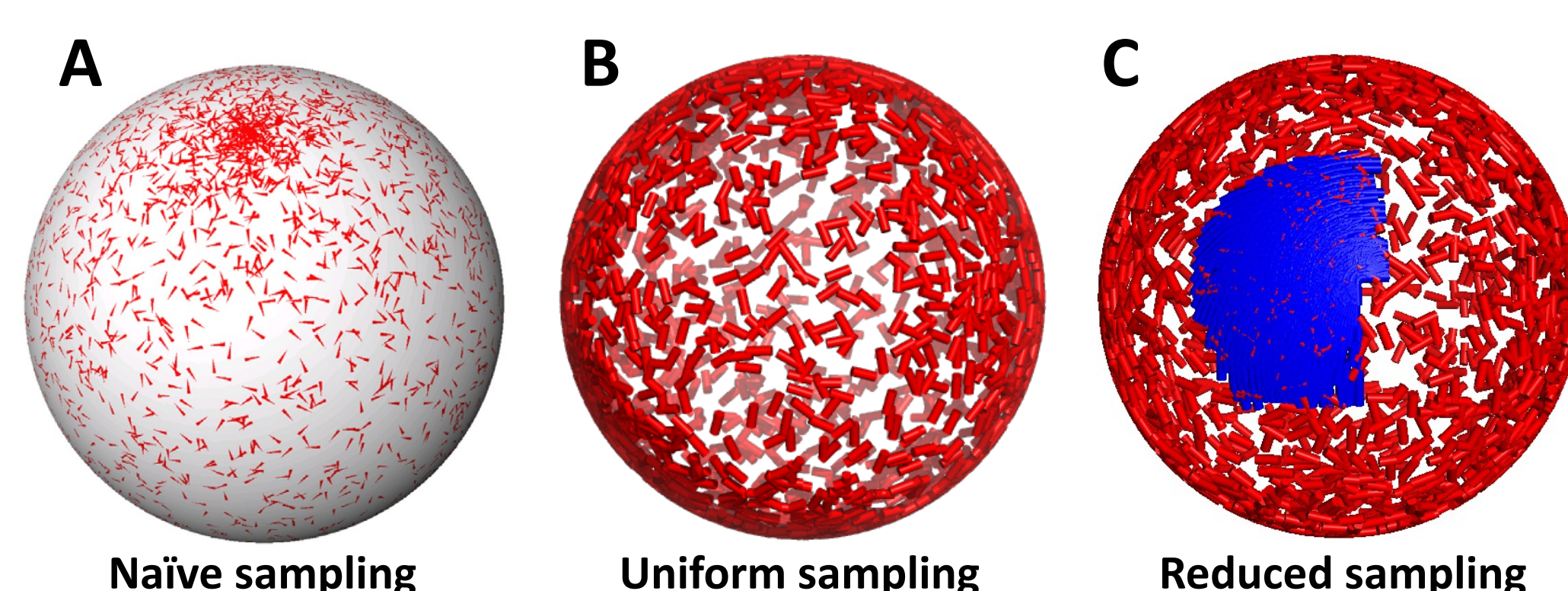


Figure 1: Sampled rotations on the globe. A) Naïvely sampling causes a bias towards the poles [5] B) The new implementation of sampling corrects this bias [5] C) Uniformly sampled rotations for TCR with respect to pMHC. In blue the restricted rotations used to dock the TCR to known binding angles to pMHC.

Results

Our approach is 10-30 times faster than Cluspro on CPU and yields models with L-RMSD < 5 Å, compared to the bound reference PDB structure [5]. A combination of rotation and distance restraints improves model quality and reduces computing time (Figure 2). The sampling quality of the 1000 candidate complexes improved (Figure 3).

Figure 2: Best model quality and computing time results for PDB ID: 1AO7 for different docking experiments. The dotted line indicates the CAPRI threshold to distinguish between acceptable (L-RMSD < 10 Å) and incorrect models (L-RMSD > 10 Å).

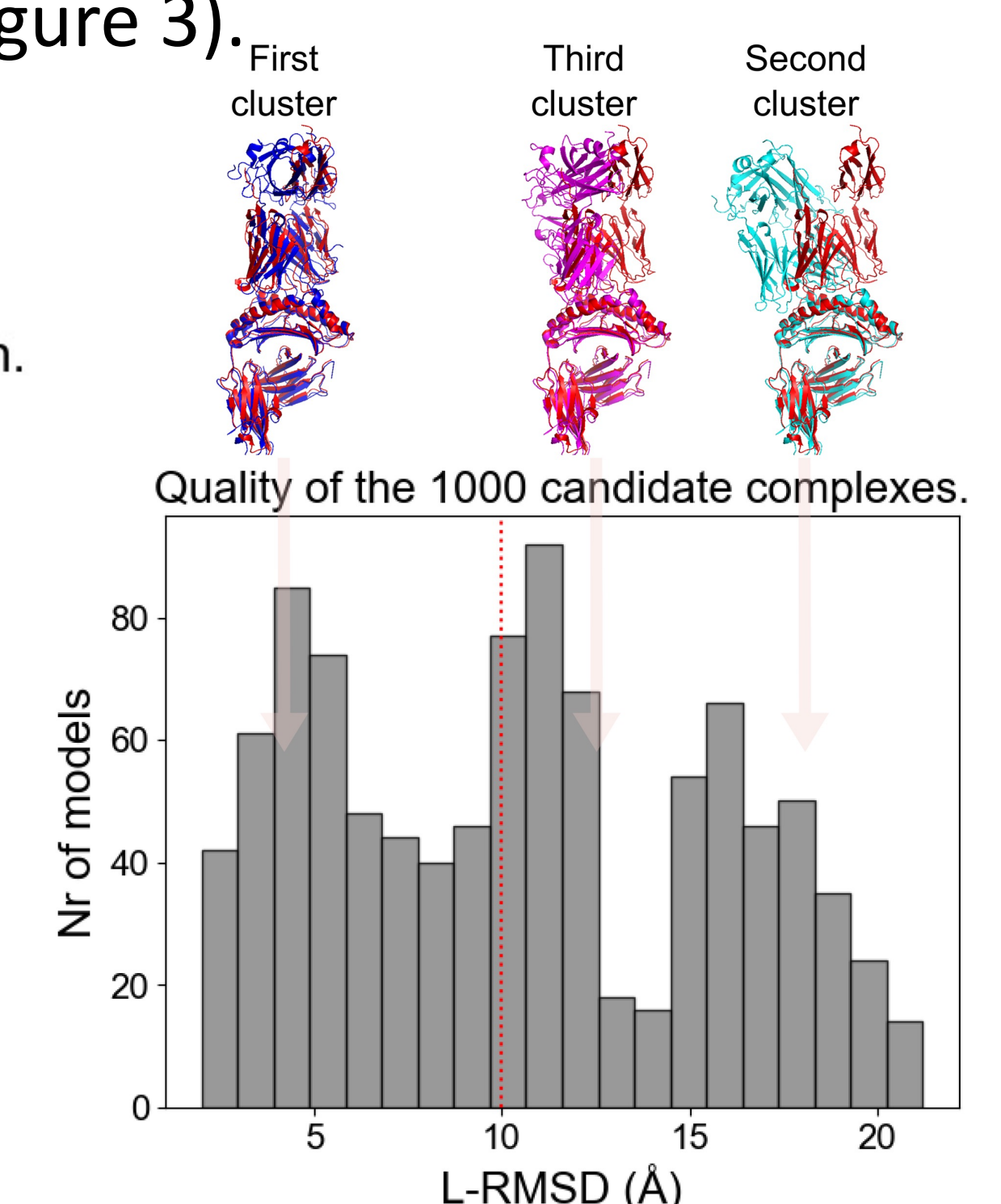
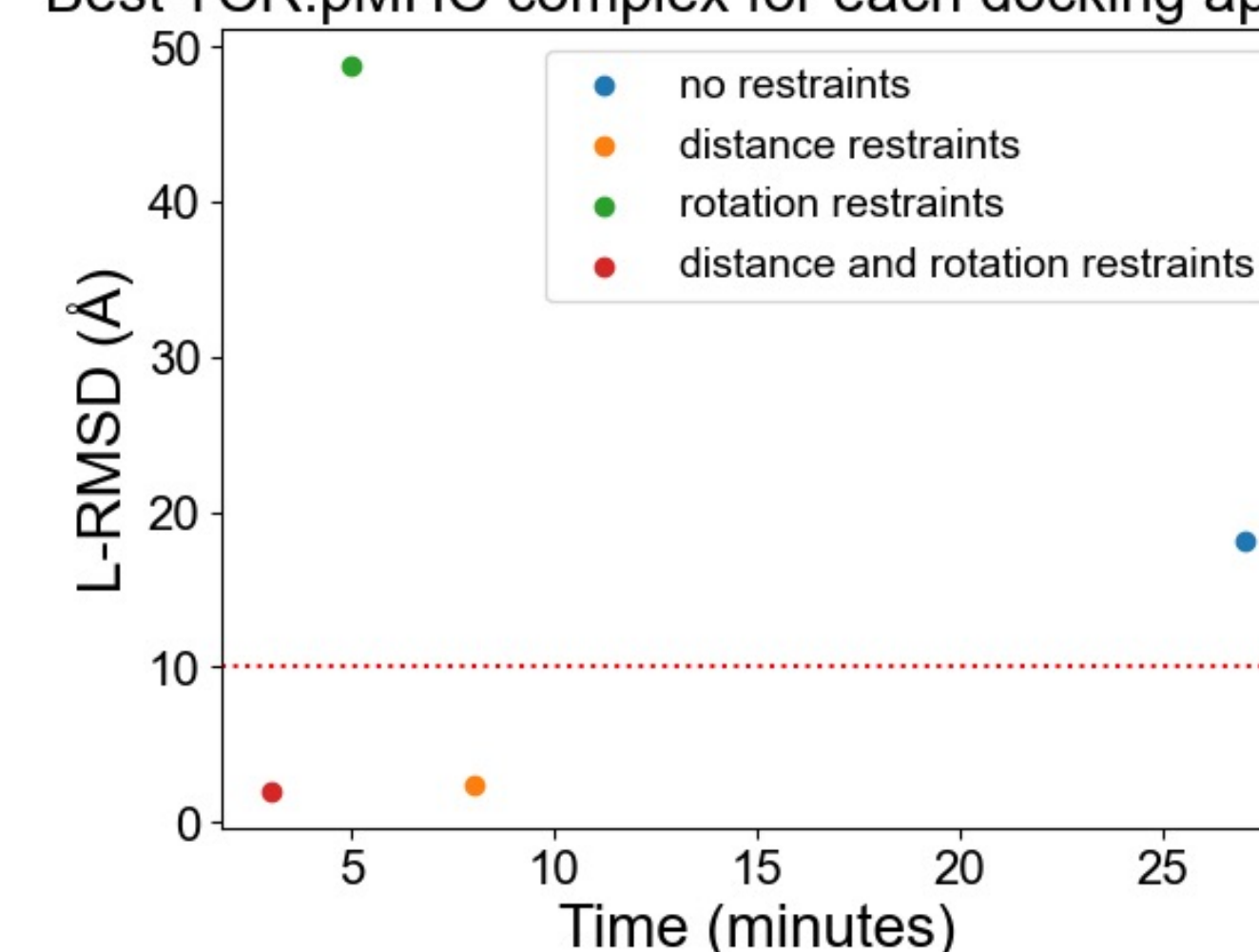


Figure 3: Sampling performance, a large number of high quality models sampled using our approach. The model results for PDB ID: 1AO7 representing the first, second and third clusters are shown in blue, magenta, cyan respectively compared to the reference structure in red.

Conclusion

- A combination of distance and rotation restraints improves model quality and at the same time reduces computing time by reducing the conformational search space for TCR:pMHC binding.
- The efficient implementation of clustering of TCR:pMHC complexes yielded to rank the near-native conformation in the first cluster.
- This approach can facilitate the large-scale modeling of TCR:pMHC complexes to generate structure-enriched training data for Deep Learning algorithms.

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I am a Bioinformatics Master Student with a focus on algorithms and Bioinformatics.

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References

1. Kozakov, Dima, et al. "PIPER: an FFT-based protein docking program with pairwise potentials." *Proteins: Structure, Function, and Bioinformatics* 65.2 (2006): 392-406.
2. Peacock, Thomas, and Benny Chain. "Information-driven docking for TCR-pMHC complex prediction." *Frontiers in Immunology* 12 (2021): 686127.
3. Kuffner, James J. "Effective sampling and distance metrics for 3D rigid body path planning." *IEEE International Conference on Robotics and Automation*, 4 (2004): 3993-3998.
4. Rudolph, Markus G., Robyn L. Stanfield, and Ian A. Wilson. "How TCRs bind MHCs, peptides, and coreceptors." *Annu. Rev. Immunol.* 24 (2006): 419-466.
5. Kozakov, Dima, et al. "The ClusPro web server for protein-protein docking." *Nature protocols* 12.2 (2017): 255-278.